

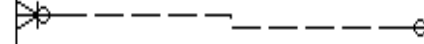
Dictionnaire des données

	Nom des colonnes	Type de données	Taille	Clé	Description
CONTRAT.CSV	Contrat_ID	INT		Clé primaire	Id unique pour les contrats
	No_voie	INT			Numéro dans la voie pour l'adresse du logement assuré
	B_T_Q	VARCHAR	1		Indicateur éventuel de répétition pour l'adresse du logement assuré sur un caractère
	Type_de_voie	VARCHAR	10		Type de voie pour l'adresse du logement assuré: rue, av (Avenue), rte (Route), ...
	Voie	VARCHAR	255		Libellé de la voie pour l'adresse du logement assuré
	Code_dep_code_commune	VARCHAR	10	Clé secondaire	Concaténation du code département et code commune pour avoir une clé unique
	Code_postal	VARCHAR	6		Code postal pour l'adresse du logement assuré
	Surface	INT			Surface de local assuré
	Type_local	VARCHAR	20		Type de local assuré
	Occupation	VARCHAR	20		Type d'occupation de local assuré
	Type_contrat	VARCHAR	20		Type de contrat de local assuré
	Formule	VARCHAR	10		Formule de cotrat d'assurance choisie
	Valeur_declaree_biens	VARCHAR	100		La valeur de bien assuré
REGION.CSV	Prix_cotisation_mensuel	INT			montant a verser chaque mois par l'assuré
	Code_dep_code_commune	VARCHAR	10	Clé primaire	Concaténation du code département et code commune pour avoir une clé unique
	reg_code	VARCHAR	2		Code le la région pour l'adresse du local assuré
	reg_nom	VARCHAR	50		Nom de la région pour l'adresse du local assuré
	aca_nom	VARCHAR	50		Nom de la capital de département pour l'adresse du local assuré
	dep_nom	VARCHAR	50		Nom de département pour l'adresse du local assuré
	com_nom_maj_court	VARCHAR	50		Nom de la commune de l'adresse du local assuré
	dep_code	VARCHAR	5		Code de département de l'adresse du local assuré
	dep_nom_num	VARCHAR	50		Concaténation du dep_nom et dep_code

Chemin relationnel

Contrat	
Contrat_ID: INTEGER	NOT NULL [PK]
No_voie: INTEGER	
B_T_Q: VARCHAR(1)	
Type_de_voie: VARCHAR(10)	
Voie: VARCHAR(255)	NOT NULL
Code_dep_code_commune: VARCHAR(10)	NOT NULL [FK]
Code_postal: VARCHAR(6)	NOT NULL
Surface: INTEGER	NOT NULL
Type_local: VARCHAR(20)	NOT NULL
Occupation: VARCHAR(20)	NOT NULL
Type_contrat: VARCHAR(20)	NOT NULL
Formule: VARCHAR(10)	NOT NULL
Valeur_declaree_biens: VARCHAR(100)	NOT NULL
Prix_cotisation_mensuel: INTEGER	NOT NULL

Region	
Code_dep_code_commune: VARCHAR(10)	NOT NULL [PK]
reg_code: VARCHAR(2)	NOT NULL
reg_nom: VARCHAR(50)	NOT NULL
aca_nom: VARCHAR(50)	NOT NULL
dep_nom: VARCHAR(50)	NOT NULL
com_nom_maj_court: VARCHAR(50)	NOT NULL
dep_code: VARCHAR(5)	NOT NULL
dep_nom_num: VARCHAR(50)	NOT NULL



- le **code SQL** générant les tables

```
CREATE TABLE Region (  
    Code_dep_code_commune VARCHAR(10) NOT NULL,  
    reg_code VARCHAR(2) NOT NULL,  
    reg_nom VARCHAR(50) NOT NULL,  
    aca_nom VARCHAR(50) NOT NULL,  
    dep_nom VARCHAR(50) NOT NULL,  
    com_nom_maj_court VARCHAR(50) NOT NULL,  
    dep_code VARCHAR(5) NOT NULL,  
    dep_nom_num VARCHAR(50) NOT NULL,  
    CONSTRAINT Region_pk PRIMARY KEY (Code_dep_code_commune)  
);
```

```
CREATE TABLE Contrat (  
    Contrat_ID INTEGER NOT NULL,  
    No_voie INTEGER,  
    B_T_Q VARCHAR(1),  
    Type_de_voie VARCHAR(10),  
    Voie VARCHAR(255) NOT NULL,  
    Code_dep_code_commune VARCHAR(10) NOT NULL,  
    Code_postal VARCHAR(6) NOT NULL,  
    Surface INTEGER NOT NULL,  
    Type_local VARCHAR(20) NOT NULL,  
    Occupation VARCHAR(20) NOT NULL,  
    Type_contrat VARCHAR(20) NOT NULL,  
    Formule VARCHAR(10) NOT NULL,  
    Valeur_declaree_biens VARCHAR(100) NOT NULL,  
    Prix_cotisation_mensuel INTEGER NOT NULL,  
    CONSTRAINT Contrat_pk PRIMARY KEY (Contrat_ID)  
);
```

```
ALTER TABLE Contrat ADD CONSTRAINT Region_Contrat_fk  
FOREIGN KEY (Code_dep_code_commune)  
REFERENCES Region (Code_dep_code_commune)  
ON DELETE NO ACTION  
ON UPDATE NO ACTION  
NOT DEFERRABLE;
```

Table contrat

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

exercise

Tables

contrat

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Administration Schemas

Information

Table: **contrat**

Columns:

<u>Contrat_ID</u>	int PK
No_voie	int
B_T_Q	varchar(1)
Type_de_voie	varchar(10)
Voie	varchar(255)
Code_dep_code_commune	varchar(10)
Code_postal	varchar(6)
Surface	int

Object Info Session

Contrat contrat contrat x

Limit to 1000 rows

```
1 • SELECT count(*) FROM contrat;
```

Result Grid

count(*)
30335

Filter Rows: Export: Wrap Cell Content: I A

Result 2 x

Output

Table region

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

exercise

Tables

contrat

region

Columns

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

sys

Contrat contrat contrat contrat x

Limit to 1000 rows

1 • `SELECT count(*) FROM region;`

Result Grid

Filter Rows: Export: Wrap Cell Content: `⌘`

count(*)
38916

Administration Schemas

Information

Table: region

Columns:

<u>Code_dep_code_commune</u>	varchar(10) PK
reg_code	varchar(2)
reg_nom	varchar(50)
aca_nom	varchar(50)
dep_nom	varchar(50)
com_nom_maj_court	varchar(50)
dep_code	varchar(5)

Object Info Session

Result 2 x Read Only

Output

Requête 1 : Lister les numéros de contrats (contrat_ID) avec leur surface pour la commune de Caen.

The screenshot shows a database query tool interface. At the top, there are tabs labeled 'Contrat', 'contrat', 'contrat', and 'contrat'. Below the tabs is a toolbar with various icons and a 'Limit to 1000 rows' dropdown. The main area contains a SQL query:

```
1 • select contrat.contrat_ID, contrat.Surface, region.com_nom_maj_court FROM contrat inner join region
2 on contrat.Code_dep_code_commune = region.Code_dep_code_commune
3 where region.com_nom_maj_court = 'CAEN';
```

Below the query, there is a 'Result Grid' section. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The result grid displays the following data:

	contrat_ID	Surface	com_nom_maj_court
▶	103791	35	CAEN
	103792	99	CAEN
	103793	40	CAEN
	103794	20	CAEN

On the right side, there is a vertical toolbar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, there is a 'Result 4' tab and an 'Output' section with a dropdown menu set to 'Action Output'. A 'Read Only' indicator is visible in the bottom right corner.

Requête 3 : Lister
le nom des régions
de France.

contrat x

Limit to 1000 rows

```
1 select distinct reg_nom from region;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

reg_nom
Grand Est
Auvergne-Rhône-Alpes
Occitanie
Provence-Alpes-Côte d'Azur
Normandie
Nouvelle-Aquitaine
Centre-Val de Loire
Hauts-de-France
Bourgogne-Franche-Comté
Bretagne
Corse

region 1 x

Read Only

Requête 4 : Quels
sont les 5
contrats qui
ont les
surfaces les
plus élevées ?

The screenshot shows a database query tool interface. At the top, a toolbar contains various icons for file operations, execution, and settings. Below the toolbar, a text area displays a SQL query:

```
1 • select contrat_ID, surface from contrat
2   order by surface DESC
3   limit 5;
```

Below the query editor, a section labeled "Result Grid" contains a table with the following data:

	contrat_ID	surface
▶	104211	815
	105463	742
	130878	595
	100822	570
	109872	559

At the bottom of the interface, a status bar shows "contrat 14" with a close icon, and a "Read Only" indicator with an information icon.

Requête 5 :
Quel est le
prix moyen
de la
cotisation
mensuelle ?

The screenshot shows a database query tool interface. At the top, there are tabs labeled 'Contrat', 'contrat', 'contrat', and 'contrat'. Below the tabs is a toolbar with various icons and a dropdown menu set to 'Limit to 1000 rows'. The main area contains a SQL query:

```
1 • SELECT avg(Prix_cotisation_mensuel) FROM `contrat`;  
2
```

Below the query editor, there is a 'Result Grid' section. It includes a 'Filter Rows:' input field, an 'Export:' button, and a 'Wrap Cell Content:' checkbox. The result grid displays the following data:

avg(Prix_cotisation_mensuel)
19.3287

On the right side of the interface, there is a vertical toolbar with icons for 'Result Grid', 'Form Editor', 'Field Types', and 'Query Stats'. At the bottom right, there is a 'Read Only' status indicator.

Requête 6 : Quel
est le nombre de
contrats pour
chaque catégorie
de prix de la valeur
déclarée des biens
?

contrat x

Limit to 1000 rows

```
1 • select Valeur_declaree_biens,  
2     count(*) as nb_contrats  
3     from contrat  
4     group by Valeur_declaree_biens;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Valeur_declaree_biens	nb_contrats
▶	0-25000	22720
	25000-50000	6815
	100000+	104
	50000-100000	696

Result 2 x

Read Only

Result Grid
Form Editor
Field Types

Requête 7 : Quel
est le nombre
de formules
“integral” sur
la région Pays
de la Loire ?

The screenshot shows a SQL query editor window with a tab labeled 'contrat'. The query is as follows:

```
1 • select
2     contrat.Formule, region.reg_nom,
3     count(*)
4 from contrat join region
5 on contrat.Code_dep_code_commune= region.Code_dep_code_commune
6 where formule = 'Integral' and reg_nom = 'Pays de la Loire'
7 group by formule,reg_nom;
```

Below the query editor, the 'Result Grid' is displayed. It shows a table with the following data:

Formule	reg_nom	count(*)
Integral	Pays de la Loire	589

The interface includes various toolbars at the top and bottom, and a sidebar on the right with options like 'Result Grid', 'Form Editor', and 'Field Types'. The bottom status bar indicates 'Result 6' and 'Read Only'.

Requête 8 : Lister les numéros de contrats avec le type de contrat et leur formule pour les maisons du département 71.

Limit to 1000 rows

```
1 • select contrat.Contrat_ID, contrat.Type_contrat, contrat.Formule, region.dep_code from contrat join region
2 ON contrat.Code_dep_code_commune= region.Code_dep_code_commune
3 where Type_local = 'Maison' and dep_code= 71;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Contrat_ID	Type_contrat	Formule	dep_code
▶	114768	Residence principale	Integral	71
	114779	Residence principale	Classique	71
	114782	Residence principale	Classique	71
	114812	Residence principale	Integral	71

Result Grid
Form Editor
Field Types

Requête 9 :
Quelle est la
surface
moyenne
des contrats
à Paris ?

The screenshot shows a SQL query editor window with a toolbar at the top containing icons for file operations, execution, and navigation. The query text is as follows:

```
1 • select region.aca_nom, avg(surface)
2   from contrat join region on
3   contrat.Code_dep_code_commune= region.Code_dep_code_commune
4   where aca_nom = 'Paris'
5   group by aca_nom;
```

Below the query editor is the 'Result Grid' window. It has a toolbar with 'Filter Rows', 'Export', and 'Wrap Cell Content' options. The results are displayed in a table with two columns: 'aca_nom' and 'avg(surface)'. There is one row for 'Paris' with a value of 51.7695.

aca_nom	avg(surface)
Paris	51.7695

On the right side of the interface is a vertical sidebar with icons for 'Result Grid' (active), 'Form Editor', 'Field Types', and 'Query Stats'. At the bottom left, it says 'Result 4 x'. At the bottom right, there is a 'Read Only' status indicator.

Requête 10 :
Classement des
10 départements
où le prix moyen
de la cotisation
est le plus élevé.

The screenshot shows a database query editor window with a tab labeled 'contrat'. The query is as follows:

```
1 • select region.dep_nom,  
2   avg (prix_cotisation_mensuel) as moy_cotisation  
3   from contrat join region  
4   on contrat.Code_dep_code_commune=region.Code_dep_code_commune  
5   group by dep_nom  
6   order by moy_cotisation desc  
7   limit 10;
```

Below the query editor is the 'Result Grid' showing the results of the query. The grid has two columns: 'dep_nom' and 'moy_cotisation'. The results are sorted in descending order of 'moy_cotisation'.

dep_nom	moy_cotisation
Paris	36.3991
Hauts-de-Seine	26.2694
Val-de-Marne	19.8242
Yvelines	18.8881
Rhône	18.4855
Ain	18.2389
Alpes-Maritimes	18.1376
Charente-Maritime	17.3173
Haute-Savoie	17.1487
Corse-du-Sud	17.0744

The interface includes various toolbars at the top and bottom, including a 'Limit to 1000 rows' dropdown, 'Export', 'Wrap Cell Content', 'Fetch rows', and 'Form Editor' buttons. The bottom status bar shows 'Result 1' and 'Read Only'.

Requête 11 :
Liste des
communes
ayant eu au
moins 150
contrats.

The screenshot shows a SQL query editor window with the following query:

```
1 • select aca_nom , count(distinct contrat_id) contrat_id
2 from contrat join region
3 on contrat.Code_dep_code_commune=region.Code_dep_code_commune
4 group by aca_nom
5 having Contrat_id>=150
```

Below the query editor is the 'Result Grid' showing the results of the query. The grid has two columns: 'aca_nom' and 'contrat_id'. The results are as follows:

aca_nom	contrat_id
Aix-Marseille	1261
Amiens	529
Besançon	181
Bordeaux	1765
Corse	247
Créteil	3996
Grenoble	1211
Lille	660
Lyon	1762
Montpellier	1136
Nancy-Metz	408

The interface includes a toolbar at the top with icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Result Grid' button on the right. The status bar at the bottom indicates 'Result 8' and 'Read Only'.

Requête 12 :
Quel est le
nombre de
contrats
pour chaque
région ?

The screenshot shows a database query editor with a query window and a results window.

Query Window:

```
1 • select reg_nom , count(distinct contrat_id) contrat_id
2   from contrat join region
3   on contrat.Code_dep_code_commune=region.Code_dep_code_commune
4  group by reg_nom;
```

Results Window:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

reg_nom	contrat_id
Auvergne-Rhône-Alpes	3042
Bourgogne-Franche-Comté	293
Bretagne	947
Centre-Val de Loire	598
Corse	247
Grand Est	769
Guyane	37
Hauts-de-France	1189
Ile-de-France	14177
La Réunion	8
Martinique	73
Normandie	824
Nouvelle-Aquitaine	2038
Occitanie	1609

Result 9 x Read Only